

Project Design Phase-II Technology Stack (Architecture & Stack)

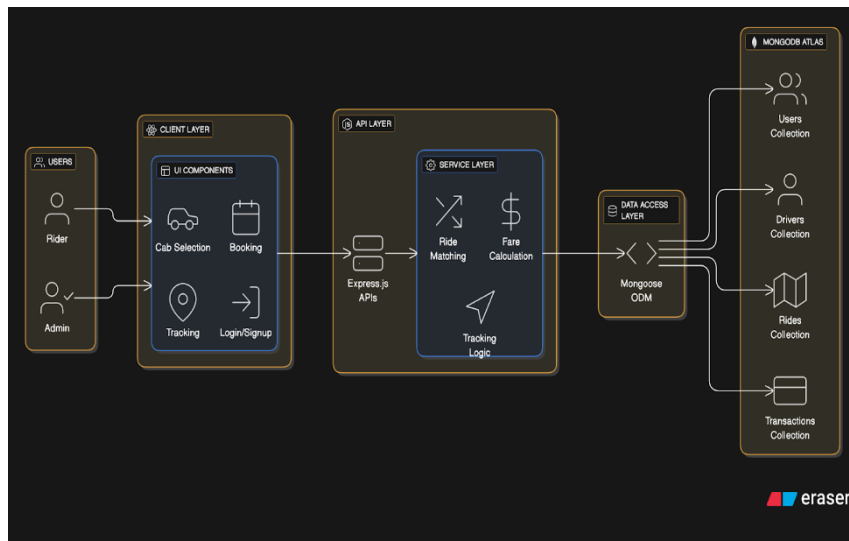
Date	6 July 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	4 Marks

Technical Architecture:

The **Cab Booking System** follows a **MERN Stack architecture**, where the React.js frontend communicates with the Node.js and Express.js backend through REST APIs. The backend manages user authentication, ride booking, driver assignment, fare calculation, and payment processing, while **MongoDB Atlas** securely stores user, driver, ride, and payment data. External services such as **Google Maps API** are integrated for location tracking and route navigation. The technical architecture diagram and the information in **Table 1** and **Table 2** illustrate the application's components, technologies, infrastructure, and data flow.

Example: Online Cab Booking and Ride Management System

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include the complete application workflow, infrastructure (Local/Cloud), external APIs (Google Maps, Payment Gateway), and MongoDB database components in the solution architecture.
- Clearly illustrate the interaction between the frontend, backend, database, and third-party services; machine learning is not applicable in the current version.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	User Interface	Web application for Riders, Drivers, and Admin	HTML5, CSS3, JavaScript, React.js
2. 2	Application Logic-1	User Authentication & Authorization	Node.js, Express.js, JWT, bcrypt
3. 3	Application Logic-2	Ride Booking & Driver Management	Node.js, Express.js
4. 4	Application Logic-3	Fare Calculation & Ride Tracking	JavaScript, Express.js
5. 5	Database	Stores Users, Drivers, Rides & Payments	MongoDB
6. 6	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7. 7	File Storage	Profile images and documents (optional)	Local Storage / Cloudinary
8. 8	External API-1	Maps, Location & Directions	Google Maps API
9. 9	External API-2	Payment Gateway	Razorpay / Stripe
10. 1 0	Machine Learning Model	Not Applicable	N/A
11. 1 1	Infrastructure (Server/Cloud)	Backend API Hosting	Node.js Server, Render / Railway / Localhost

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Secure Login, Password Encryption & JWT Authentication	bcrypt, JWT, HTTPS
3.	Scalable Architecture	MERN Stack supports future growth and multiple users	React.js, Express.js, MongoDB
4.	Availability	Application available 24×7 through cloud deployment	MongoDB Atlas, Render
5.	Performance	Fast API response and optimized database queries	Express.js, MongoDB Indexing

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>